Riemann Sums and Definite Integrals

ANSWER KEY

Section 1: Left and Right Riemann Sums

1 Approximate $\int_0^4 x^2\,dx$ using a Left Riemann sum with $n = 4$ equal subintervals.

a. 14 ✓

b. 30

c. 21

d. 16

$\Delta x = \frac{4-0}{4} = 1$. Grid points: $0, 1, 2, 3, 4$. Left sum
$= 1 \cdot [f(0) + f(1) + f(2) + f(3)] = 1 \cdot [0 + 1 + 4 + 9] = 14$.

2 Approximate $\int_0^4 x^2\,dx$ using a Right Riemann sum with $n = 4$ equal subintervals.

a. 30 ✓

b. 14

c. 22

d. 28

Right sum $= 1 \cdot [f(1) + f(2) + f(3) + f(4)] = 1 \cdot [1 + 4 + 9 + 16] = 30$.

3 For a strictly increasing function $f(x)$ on $[a, b]$, which inequality correctly relates the exact integral I to the Left (L_n) and Right (R_n) Riemann sums?

a. $L_n < I < R_n$ ✓

b. $R_n < I < L_n$

c. $I < L_n < R_n$

d. $L_n = I = R_n$

For an increasing function, rectangles evaluated at the left endpoints lie below the curve ($L_n < I$) and right endpoints lie above the curve ($I < R_n$).

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Section 2: Midpoint and Trapezoidal Approximations

4 Approximate $\int_1^5 \frac{1}{x}\,dx$ using the Midpoint Rule with $n = 2$ equal subintervals.

a. $\frac{1}{2} + \frac{1}{4} = 0.75 \approx 1.5$ **is wrong; width=2:** $2(1/2 + 1/4) = 1.5$ ✓

b. 1.5

c. 1.2

d. 1.33

$\Delta x = \frac{5-1}{2} = 2$. Subintervals: $[1, 3]$ and $[3, 5]$. Midpoints are $m_1 = 2$ and $m_2 = 4$. Midpoint sum $= 2 \cdot [f(2) + f(4)] = 2(1/2 + 1/4) = 2(3/4) = 1.5$.

5 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^2 x^3\,dx$.

a. 4.25 ✓

b. 4.00

c. 3.75

d. 4.50

$\Delta x = 0.5$. Grid: $0, 0.5, 1, 1.5, 2$. Values: $0, 0.125, 1, 3.375, 8$.
$T_4 = \frac{0.5}{2}[0 + 2(0.125 + 1 + 3.375) + 8] = 0.25[0 + 9 + 8] = 0.25(17) = 4.25$.

6 If $f''(x) > 0$ (concave up) on $[a, b]$, how does the Trapezoidal approximation T_n compare to the exact integral I?

a. $T_n > I$ **(overestimate)** ✓

b. $T_n < I$ (underestimate)

c. $T_n = I$

d. Cannot be determined without $f'(x)$

When f is concave up, secant lines connecting adjacent points on the curve lie entirely above the curve, so trapezoids overestimate the area: $T_n > I$.

Section 3: Definite Integral as the Limit of a Riemann Sum

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7 Express the limit $\lim_{n\to\infty}\sum_{i=1}^{n}\left(2+\frac{3i}{n}\right)^4\frac{3}{n}$ as a definite integral.

a. $\int_2^5 x^4\,dx$ ✓

b. $\int_0^3 (2+x)^4\,dx$

c. $\int_2^5 (2+3x)^4\,dx$

d. $\int_0^1 (2+3x)^4\,dx$

Here $\Delta x = 3/n$ and $x_i = 2 + i\Delta x$. Lower limit $a = 2$, upper limit $b = 2+3 = 5$. Integrand is x^4. Integral is $\int_2^5 x^4\,dx$.

8 Evaluate $\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{i^2}{n^2}$.

a. $1/3$ ✓

b. $1/2$

c. $1/4$

d. 1

This corresponds to $\int_0^1 x^2\,dx = [x^3/3]_0^1 = 1/3$.

9 Convert $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{1}{n+i}$ into a definite integral on $[0,1]$ and evaluate it.

a. $\ln 2$ ✓

b. 1

c. $\ln 3$

d. $1/2$

Factor out $1/n$: $\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+i/n} = \int_0^1 \frac{1}{1+x}\,dx = [\ln(1+x)]_0^1 = \ln 2$.

Section 4: Geometric Evaluation of Definite Integrals

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10 Evaluate $\int_{-3}^{3} \sqrt{9 - x^2}\, dx$ using geometry.

a. $\frac{9\pi}{2}$ ✓

b. 9π

c. $\frac{9\pi}{4}$

d. 18

The graph of $y = \sqrt{9 - x^2}$ is the upper semicircle of radius $r = 3$ centered at the origin. Area $= \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(9) = \frac{9\pi}{2}$.

11 Evaluate $\int_{0}^{4} (2x - 3)\, dx$ by computing net signed area.

a. 4 ✓

b. 8

c. 0

d. -2

Triangles: below axis on $[0, 1.5]$: base 1.5, height 3, area $-(1.5 \times 3)/2 = -2.25$. Above axis on $[1.5, 4]$: base 2.5, height 5, area $(2.5 \times 5)/2 = 6.25$. Net $= 6.25 - 2.25 = 4$.

12 If $\int_{1}^{5} f(x)\, dx = 7$ and $\int_{3}^{5} f(x)\, dx = 4$, find $\int_{1}^{3} f(x)\, dx$.

a. 3 ✓

b. 11

c. -3

d. $7/4$

By additivity of intervals: $\int_{1}^{5} f = \int_{1}^{3} f + \int_{3}^{5} f \Rightarrow 7 = \int_{1}^{3} f + 4 \Rightarrow \int_{1}^{3} f = 3$.

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